

MobilityHexVis: Analyzing socioeconomic urban mobility patterns through small multiples visualization

EPB: Urban Analytics and City Science

2026, Vol. 0(0) 1–17

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DOI: 10.1177/23998083261437300

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Abstract

Urban mobility patterns reveal important socio-spatial phenomena, yet visualizing these patterns from massive datasets presents significant challenges. We introduce MobilityHexVis, an interactive visualization platform for analyzing urban mobility flows from mobile phone data classified by socioeconomic status. Our approach employs small multiples visualization with hexagonal glyphs, enabling simultaneous comparison of mobility patterns across different socioeconomic groups and distance ranges. Each hexagon displays origin-destination proportions using a bi-color split representation, while a Gini index quantifies social mixture in urban areas. Through a case study in Santiago, Chile, we demonstrate how the platform reveals employment centers and socioeconomic differences in mobility patterns. A pilot study confirms the platform's effectiveness while identifying areas for improvement. MobilityHexVis contributes to urban mobility understanding and provides a tool to support evidence-based planning toward more inclusive cities.

Keywords

urban mobility, socioeconomic segregation, geovisualization, small multiples, mobile phone data

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Data Availability Statement included at the end of the article

Introduction

Understanding urban mobility patterns is essential for analyzing complex socio-spatial phenomena in modern cities. As urban areas grow and change demographically, the analysis of population movement flows offers insights into how residents interact with their environment (Yu et al., 2025). Effective dialogue tools and visualization approaches are increasingly needed to support communication in transdisciplinary urban planning environments (Billger et al., 2017).

Urban mobility, defined as the sum of individual movements that allow citizens to access labor markets, goods, and services (Brůhová Foltýnová et al., 2020), has been traditionally studied through surveys. While these methods provide detailed demographic information, they face practical limitations such as high costs, lengthy execution times, and potential biases due to limited sample sizes (Dios Ortúzar and Willumsen, 2021). These constraints make it difficult to capture the dynamic complexity of urban mobility and delay the acquisition of updated information, limiting their usefulness for responding to emerging trends (Demissie et al., 2013).

To overcome these limitations, traditional data sources are increasingly being complemented with new sources that offer different advantages. In the last decade, massive data obtained from mobile phones and GPS have transformed the study of urban mobility. These sources collect information continuously and passively, offering superior spatial and temporal resolution compared to traditional surveys (Osorio-Arjona and García-Palomares, 2019). They allow for dynamic analysis of socio-spatial patterns and represent an opportunity to study urban mobility more accurately and consistently (Müürisepp et al., 2022). These data not only enhance our understanding of mobility patterns but can also be used to define functional urban boundaries based on actual movement patterns, overcoming limitations of administrative boundaries (Dong et al., 2024; Sotomayor-Gómez and Samaniego, 2020).

The ability to visualize movement flows intuitively is essential for identifying, studying, and communicating mobility patterns, which can be used to understand various socio-spatial phenomena and support decision-making. However, visualizing high-volume flows is notoriously difficult due to the high density of flow lines that cause visual clutter (Graser et al., 2019). This challenge is particularly relevant when studying social segregation or mixing, concepts associated with the degree of social diversity in a given geographical area (Ruiz-Tagle and Romano, 2019), which has become an urban policy objective in many regions, including Chile (Vergara, 2021).

While platforms have been developed for the analysis of geospatial data from mobile phones or GPS, there is limited work specifically focused on comparing movement patterns across different socioeconomic groups. Zhou et al. (2018) developed a comprehensive system for taxi movement visualization using hexagonal partitioning and matrix factorization, but their work focuses on identifying urban functional areas rather than socioeconomic comparisons. Our work builds upon their hexagonal method by innovatively integrating simultaneous socioeconomic group comparison through small multiples. This unique approach fills a critical gap, positioning our proposal as a novel solution. We present MobilityHexVis, an interactive visualization platform designed for the analysis of urban mobility flows, leveraging comprehensive mobile phone data categorized by socioeconomic status.

Our approach uses small multiples visualization with hexagonal glyphs, allowing simultaneous comparison of mobility patterns across different socioeconomic groups and distance ranges. By separating users into socioeconomic groups based on their inferred place of residence from displacement behavior, along with socioeconomic indices, the platform facilitates exploration of phenomena such as segregation or social mixing. The system implements spatial aggregation techniques with hexagonal tessellation and employs the Gini index to quantify social mixture in urban areas.

The contributions of this research are threefold: (1) We propose a novel small multiples approach for visualizing and comparing urban mobility flows across different socioeconomic groups; (2) We demonstrate the effectiveness of hexagonal glyphs with bi-color split representation to show the proportion between origins and destinations; and (3) We provide insights on urban mobility patterns in Santiago, Chile, that highlight the segregation of socioeconomic groups using mobile phone registries during business-as-usually urban experiences.

The remainder of this paper is organized as follows: First, we review related work on urban mobility visualization and social segregation studies. Next, we describe our data and methods, followed by the system design of MobilityHexVis. We then present a case study in Santiago, Chile, demonstrating the platform's capabilities, and discuss findings from a pilot study with users. Finally, we summarize our contributions and outline directions for future work.

Related work

Our research builds upon previous work in three main areas: the use of mobile phone data for urban mobility analysis, visualization techniques for mobility flows, and studies on social segregation in urban environments.

Early work by Hägerstrand (1970) proposed Time Geography as a framework for defining social interactions spatially, which gained significant traction with new technologies that facilitate tracking individual trajectories. For instance, Farber et al. (2013) leveraged this approach to develop an integrated method for understanding the social interaction potential of places, offering crucial insights amidst the 'data deluge' generated by mobile phone technology.

Mobile phone data has become valuable for urban mobility studies due to its high spatiotemporal resolution and passive collection method (Blondel et al., 2015). These data offer advantages over traditional surveys, including continuous collection, extensive coverage, and lower costs (García-Albertos et al., 2017; Torres et al., 2021). Researchers have leveraged this data to infer transportation modes (Graells-Garrido, Opitz, et al., 2023a), identify areas of interest (Graells-Garrido, Schifanella, et al., 2023b), study everyday geographies (De Sabbata et al., 2024), understand cities functionally (Dong et al., 2024), and define urban boundaries through travel patterns (Sotomayor-Gómez and Samaniego, 2020).

Visualizing mobility flows presents unique challenges due to data complexity and volume. Flow maps are common for origin-destination (OD) data visualization (Robinson, 1967; Tobler, 1987), but can become cluttered with high-volume data (Graser et al., 2019). Alternative approaches include OD matrices (Pérez-Messina et al., 2020) and OD maps (Slingsby et al., 2014; Wood et al., 2013). Recent innovations include Pérez-Messina and Graells-Garrido (2019)'s glyph-based flow maps for studying commuting flows, Zhou et al. (2018)'s hexagonal glyphs for taxi movement patterns, and Odell et al. (2022)'s combination of network analysis with spatial visualization to identify polycentric structures. Peña-Araya et al. (2020) empirically evaluated different visualization strategies, finding that small multiple maps perform best overall for geographical phenomena.

Building upon hexagonal visualization approaches, recent work has established theoretical foundations for glyph-based spatial representations. Wickham et al. (2012) developed systematic frameworks for embedding temporal patterns in spatial contexts through coordinate transformation, while the effectiveness of hexagonal tessellation for flow visualization has been demonstrated in multiple domains (Birch et al., 2007; Sahr et al., 2003). Jacques Bertin's seminal work on graphic semiology (Bertin, 1983) provides the theoretical foundation for encoding multiple data dimensions through visual variables such as size, value, texture, color, orientation, and shape. This foundational work on the systematic use of graphic variables for visual communication, particularly in the context of maps and diagrams, is crucial for our approach to socioeconomic comparisons, which we adapt for socioeconomic comparisons.

Social segregation studies have traditionally relied on residential segregation indices (Duncan and Duncan, 1955; James and Taeuber, 1985), but recent research emphasizes considering mobility patterns as people's segregation experiences are influenced by daily movements (Shareck et al., 2014; Wong, 2016). Lenormand and Samaniego (2023) demonstrated how individuals from similar socioeconomic contexts visit similar locations and interact with people sharing their socioeconomic condition. Xu et al. (2025) showed that lower-income neighborhoods have more constrained mobility patterns, traveling shorter distances and accessing fewer amenities compared to wealthier individuals. Moro et al. (2021) used large-scale mobile phone data to reveal how mobility patterns shape experienced income segregation, demonstrating the value of combining movement data with socioeconomic analysis.

The Gini index, a widely recognized measure of statistical dispersion, is conventionally employed to quantify income or wealth inequality within a population. Its application has been extended to quantify segregation in mobility patterns (Yao et al., 2019; Östh et al., 2018), thereby providing a more comprehensive understanding of socioeconomic group experiences through mobility. Despite these advances, there remains a gap in tools that specifically focus on comparing mobility patterns across different socioeconomic groups. Existing platforms often emphasize general mobility patterns (Du et al., 2021; Wang et al., 2017) or focus on specific transportation modes (Di Lorenzo et al., 2016), but few address socioeconomic differences explicitly.

Data and methods

This section describes the data sources and methodological approaches used in developing MobilityHexVis. We first present the mobile phone dataset and how socioeconomic status was assigned to users. Then, we outline our data processing pipeline and the metrics employed to quantify social mixing in urban mobility flows.

Mobile phone data

The primary dataset used in this research consists of Extended Detail Records (XDR) from mobile phone users in Santiago, Chile, provided by Telefónica I + D Chile. The dataset covers four seven-day periods throughout 2015 (March, May, August, and November), comprising approximately 400 GB of data with nearly 3 billion records. Each record contains a timestamp, device identifier (anonymized), and the geographic coordinates of the base transceiver station (BTS) that captured the connection.

In 2015, Telefónica (commercially known as Movistar in Chile) held 35.2% of the contract mobile phone market and 37.3% of the prepaid market that year. While this representation is substantial, we acknowledge potential sampling biases. Mobile phone users may not fully represent certain demographic groups, particularly the elderly who may have lower mobile phone adoption rates, and very young children. Additionally, the spatial distribution of users across different providers may not be uniform, potentially affecting our socioeconomic classification.

Socioeconomic classification

To classify users by socioeconomic status, we used the Socio-Material and Territorial Index (ISMT) (Observatorio De Ciudades UC, 2022). This index assigns a value between 0 and 1 to each census zone based on four key variables from the 2017 census: household head's education level, housing material quality, overcrowding, and presence of extended family members sharing housing. Unlike traditional socioeconomic groupings that focus primarily on purchasing power, the ISMT provides a

more comprehensive perspective by including aspects reflecting quality of life and material environment.

We inferred each user's place of residence based on their most frequent nighttime location (between 10 PM and 6 AM) during weeknights (Monday through Thursday). By cross-referencing these locations with ISMT data, we assigned each user to one of four quartiles (Q1 to Q4, where Q4 represents the highest socioeconomic level). This approach aligns with established methodologies in mobile phone data research (Blumenstock et al., 2015; Soto et al., 2011).

The reliability of this classification depends on the homogeneity of ISMT values within census zones. While census zones in Santiago are designed to be relatively homogeneous, we recognize that some zones may contain internal socioeconomic diversity. Additionally, the temporal gap between our mobility data (2015) and the census data (2017) assumes relative stability in socioeconomic spatial patterns. This assumption is reasonable for Santiago, where residential patterns change slowly, but should be considered when interpreting results.

Data processing pipeline

Our data processing pipeline transforms raw XDR records into hexagonal origin-destination (OD) flows classified by socioeconomic status. The process includes the following key steps:

1. Trajectory reconstruction: We grouped records by user and chronologically ordered them to reconstruct hourly trajectories. For each hour of the day, we assigned the last known location, assuming static position if no records existed for that hour.
2. OD flow extraction: For each user, we identified origin and destination nodes within specified time ranges. A flow was recorded when origin and destination differed, capturing the user's movement pattern.
3. Spatial aggregation: We implemented hexagonal spatial aggregation using the H3 indexing system (Uber Technologies Inc., 2024), which divides geographical space into uniform hexagonal cells. We chose hexagonal tessellation over square grids due to its established advantages for flow visualization. Hexagons offer uniform distance to all six neighbors, eliminating the directional bias found in square grids' orthogonal versus diagonal neighbors, and provide an optimal perimeter-to-area ratio among regular tessellations (Birch et al., 2007). Each BTS location was mapped to its corresponding hexagon at various resolution levels.
4. Flow aggregation: We aggregated flows by origin hexagon, destination hexagon, and socioeconomic group, computing the distance between hexagons to enable distance-based filtering in the visualization.

The processed data was stored in Protocol Buffers format for efficient storage and retrieval in the web application. This approach significantly reduced file sizes compared to text-based formats like CSV, improving loading times and overall performance.

We acknowledge that our analysis is subject to the Modifiable Areal Unit Problem (MAUP), whereby different spatial aggregation schemes can produce different analytical results (Openshaw, 1984). While this issue is discussed by Garreton et al. (2020) in a largely social context for Santiago, we address this limitation by allowing users to dynamically adjust hexagonal resolution through the interface (Figure 1, 'Resolución hexágonos'), enabling multi-scale analysis. The H3 hierarchical structure supports multiple resolution levels (from coarse to fine-grained), allowing users to explore how patterns change across scales and partially mitigate MAUP effects through comparative analysis. We acknowledge, however, that H3's predetermined discretization at a fixed scale does not adapt to ecologically meaningful spatial units, unlike approaches that partition space based on

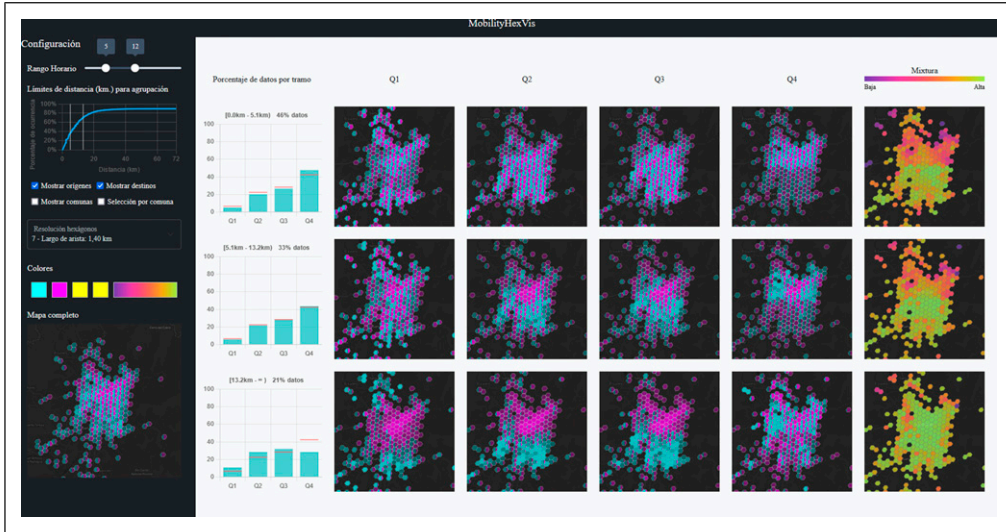


Figure 1. MobilityHexVis interface showing the configuration panel (left) and visualization matrix (right) with hexagonal flow maps for different socioeconomic groups and distance ranges. The last column of the matrix contains mixture maps showing socioeconomic diversity of flows using the Gini index.

nominal attributes (Slingsby and Loon, 2016). This is a deliberate trade-off: H3 has become a standard data exchange format in local urban analytics, which favors reproducibility and interoperability across studies.

Measuring social mixing

To quantify the degree of social mixing or segregation in mobility flows, we employed the Gini index, which measures inequality in the distribution of a resource or attribute across a population (Crespo and Hernandez, 2020). In our context, the Gini index evaluates how evenly mobility flows are distributed among different socioeconomic groups in specific urban areas.

For a set of data representing flow distribution across different socioeconomic categories, the Gini index is calculated as:

$$G = 1 - \frac{2}{n-1} \left(n - \frac{\sum_{i=1}^n \sum_{j=1}^i x_j}{\sum_{i=1}^n x_i} \right)$$

Where n is the total number of observations, and x_j represents the accumulated amount up to position j ordered from smallest to largest.

A Gini value close to 0 indicates that mobility flows are distributed homogeneously among different socioeconomic groups, suggesting a more inclusive environment. Conversely, a value closer to 1 indicates that flows are dominated by a single group, potentially signaling socioeconomic segregation in mobility.

This analytical approach allows MobilityHexVis to identify areas where certain groups experience significant differences in mobility access and patterns, which can be an indicator of structural inequality and urban segregation.

System design

MobilityHexVis is designed to support the exploration and comparison of mobility patterns across different socioeconomic groups. Following a nested model for visualization design (Munzner, 2009), we first identified analytical tasks through literature review and expert interviews, then developed visual encodings and interaction techniques to support these tasks.

Design goals

The primary goal of MobilityHexVis is to facilitate the analysis of urban mobility patterns with a focus on socioeconomic comparisons. As a visualization system, our contribution lies in enabling analyses that were previously impractical or impossible with existing tools. While platforms like those developed by Zhou et al. (2018) focus on identifying urban functional areas through taxi data, MobilityHexVis specifically addresses the gap in comparing mobility patterns across socioeconomic groups through simultaneous visualization. Based on the literature review and the characterized problem, we identified four key analytical tasks the system should support:

1. Identification of general mobility patterns: Providing a macro view of population flows between different urban areas and analyzing movement trends across different time periods.
2. Analysis of flow diversity across socioeconomic groups: Evaluating spatial integration or segregation by identifying areas with high or low socioeconomic diversity, detecting segregated areas dominated by a single group, and exploring patterns at different spatial resolutions through dynamic hexagon sizing.
3. Spatial analysis of mobility flows: Examining spatial characteristics of movements between socioeconomic groups, comparing flow intensity between areas, differentiating between short and long-distance trips, and identifying key activity centers.
4. Temporal analysis of mobility: Studying movement patterns across different time ranges, comparing peak and off-peak hours, and analyzing activities outside conventional working hours.

To address these tasks, we adopted a small multiples approach that enables simultaneous comparison of mobility patterns across different dimensions. The choice of small multiples implements a juxtaposition strategy (Gleicher et al., 2011), allowing users to make rapid comparisons across socioeconomic groups while maintaining spatial context (Peña-Araya et al., 2020). This design decision prioritizes comparative analysis over detail in individual views, aligning with our goal of revealing socioeconomic patterns. We implemented hexagonal spatial aggregation to reduce visual complexity while preserving geographical context.

System architecture

MobilityHexVis is implemented as a web application, with source code available at <https://github.com/PLUMAS-research/MobilityHexVis> under an open-source license. The application follows a client-side architecture where intensive data processing occurs during a pre-processing stage, and the client application focuses on visualization and interaction. The system uses Protocol Buffers for efficient data storage and standard web visualization libraries (Bostock et al., 2011).

Interface components

The MobilityHexVis interface consists of two main panels: a configuration panel on the left and a visualization matrix panel on the right (Figure 1). The configuration panel allows users to select time ranges, distance thresholds, and visualization options. The visualization matrix displays multiple views organized in a grid layout, with rows corresponding to distance ranges and columns to socioeconomic groups.

The primary visual element is the hexagonal flow map, which represents mobility patterns using hexagonal glyphs on a cartographic map. Each hexagon corresponds to a geographic area defined by the H3 indexing system. For each hexagon, we compute the number of occurrences as an origin (O_{H_i}) and as a destination (D_{H_i}) in the OD flows. The total activity associated with the hexagon is defined as $C_T = O_{H_i} + D_{H_i}$, and the proportions of origins and destinations are calculated as $P_O = O_{H_i}/C_T$ and $P_D = D_{H_i}/C_T$.

Each hexagonal glyph is colored using a bi-color split that represents these proportions, with cyan representing origins and magenta representing destinations (Figure 2). The split is positioned proportionally to represent the exact data values: if a hexagon has 75% origins and 25% destinations, the cyan area occupies exactly 75% of the hexagon's area. This precise visual encoding ensures accurate representation of the underlying data. The opacity of the glyph is proportional to the percentage of flows involving the hexagon relative to the total number of flows. Figure 3 illustrates how these hexagonal glyphs appear in the flow map visualization, including examples of selected hexagons and administrative boundaries. Importantly, the visualization interface allows users to customize these colors according to their preferences, ensuring that the system remains accessible to users with different visual needs and providing flexibility to adapt the visualization to various presentation contexts. The default map background uses dark-themed tiles, which increases the contrast of the cyan-magenta encoding and aligns with preferences expressed by local practitioners in urban analytics contexts; the map background is an application setting.

Each row in the visualization matrix contains a distribution chart showing the percentage of flows by socioeconomic group, four hexagonal flow maps (one for each socioeconomic quartile), and a mixture map showing the Gini index of socioeconomic distribution (Figure 1, last column to the right). The mixture map visualizes the socioeconomic diversity of flows in each hexagonal area, with darker colors indicating higher inequality (lower diversity) and lighter colors indicating lower inequality (higher diversity).

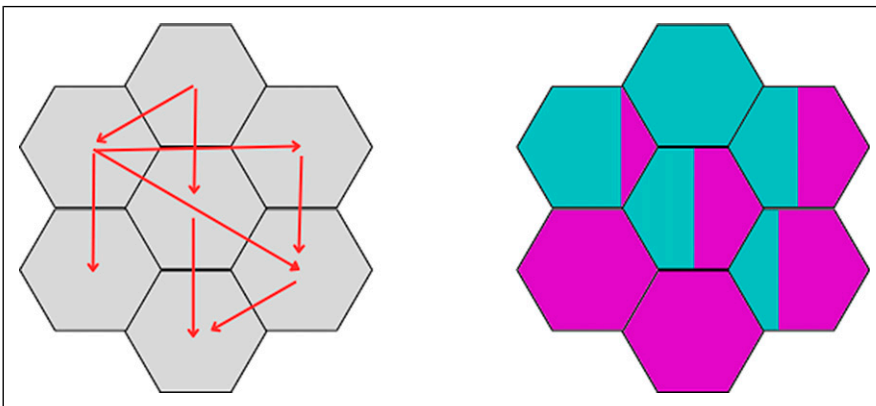


Figure 2. Representation of movement flows using hexagonal glyphs. On the left, hexagonal movement flows are shown with red arrows. On the right, the representation using hexagonal glyphs.

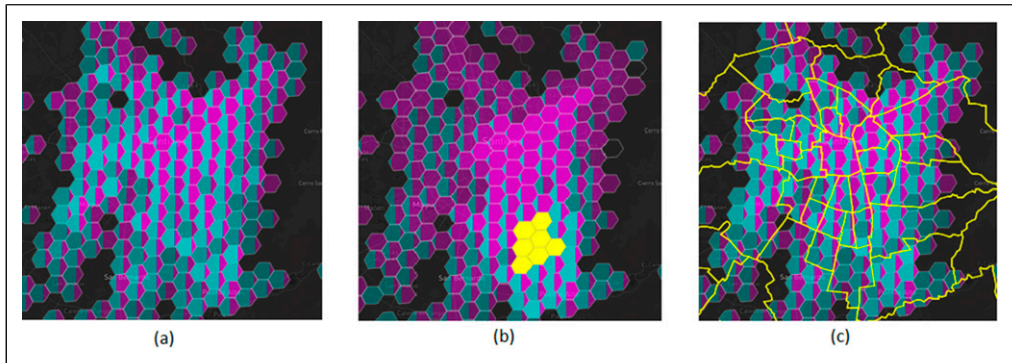


Figure 3. Hexagonal flow map visualization: (a) Hexagonal glyphs colored according to origin-destination proportions. (b) Selected hexagons highlighting specific areas. (c) Map with administrative boundaries.

Interaction mechanisms

MobilityHexVis implements the following interaction mechanisms: spatial resolution adjustment, time range selection, distance threshold filtering, hexagon selection, municipality selection, pan and zoom, and hover tooltips. A hexagon resolution dropdown (visible in Figure 1) allows users to adjust the spatial aggregation level, enabling exploration of patterns at different scales and helping assess the sensitivity of findings to the MAUP. Users can select start and end times for the mobility flows, with default values of 5:00 and 12:00, corresponding to the typical morning commute period. A cumulative percentage diagram (see Figure 1 in top-left area) shows the relationship between distance and the cumulative percentage of OD flows, allowing users to add distance thresholds that create new rows in the visualization matrix. Hovering over a hexagon displays a tooltip with the exact origin and destination counts and proportions for the selected socioeconomic group and distance range.

Users can select one or more hexagons by clicking on them, which filters the displayed flows to show only those involving the selected hexagons. A municipality selection mode allows selecting all hexagons within a specific administrative boundary with a single click, facilitating analysis at the municipal level. Pan and zoom operations are synchronized across all maps in the visualization matrix, maintaining consistent geographical context.

The system's design supports future extensibility through modular architecture. While the current implementation focuses on socioeconomic comparisons across distance ranges, the framework could be adapted to compare different temporal periods simultaneously (as suggested in our pilot study), incorporate additional demographic dimensions, or integrate other data sources. The use of Protocol Buffers and standardized visualization libraries facilitates these potential extensions.

Case study

To demonstrate the capabilities of MobilityHexVis, we conducted a case study analyzing mobility patterns in Santiago, Chile. This section presents two analyses that showcase how the platform can be used to identify employment activity centers and compare mobility patterns across different socioeconomic groups.

Identifying employment activity centers

In Chile, the typical workday and academic schedule occurs between 8:00 and 18:00 hours. Trips from place of residence to workplace or study typically occur between 5:00 and 9:00 in the morning. Using the configuration panel, we selected this time range to visualize data associated with morning commute trips.

Observing the overview map, we noted that hexagonal glyphs located in the central area of the city showed a higher proportion of magenta color (destination) compared to hexagons in the rest of the city ([Figure 4\(a\)](#)). This pattern aligns with previous research identifying a core employment area in central and northeastern Santiago, including the historic center (Santiago commune) and extending to the northeast through the communes of Providencia, Las Condes, and Vitacura ([Escolano Utrilla et al., 2015](#)).

By activating the municipality selection mode, we selected these communes with the cursor, including all hexagonal glyphs contained within them. We observed that most of the selected hexagonal glyphs showed a high proportion of cyan color (origin), indicating that individuals typically move from various areas of the city toward the selected communes ([Figure 4\(b\)](#)).

To perform a similar analysis during the time when most people return from work to their places of residence, we selected the time range from 17:00 to 21:00 hours. The pattern of colors reversed: most hexagons showed a predominance of magenta, which can be interpreted as individuals moving from the communes of interest to different sectors of the city ([Figure 4\(c\)](#)). This behavior further reinforces the notion of a core employment activity area in the central-northeastern zone of the city.

Comparing mobility patterns across socioeconomic groups

To explore how mobility patterns vary across different socioeconomic levels and how they compare in long-distance trips, we examined the cumulative percentage diagram in the configuration panel. This visualization shows how data is distributed according to travel distance. We set a distance threshold at 10 km, which created two rows in the visualization matrix panel ([Figure 5](#)). We used the 5:00 to 9:00 hour range for exploration purposes.

In the socioeconomic group distribution chart (first column), we observed a distinctive pattern for the highest socioeconomic group (Q4). For trips less than 10 km (first row), Q4's relative participation was higher compared to its baseline value (red line), while for trips greater than 10 km (second row), its participation was considerably lower. This pattern was not observed in the other

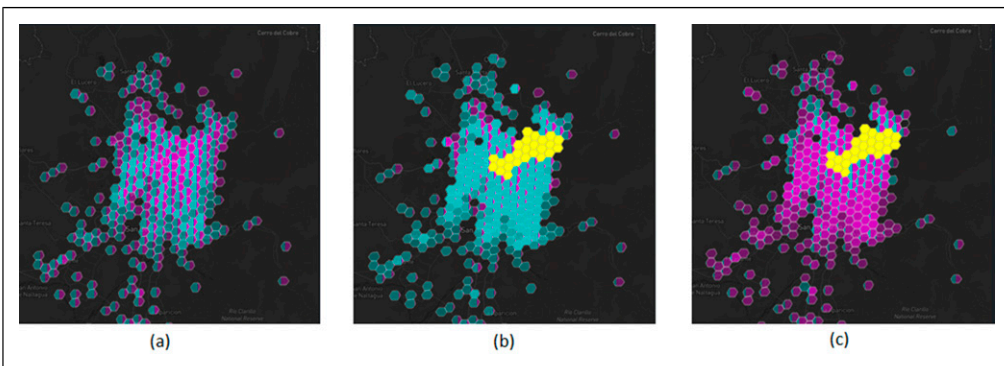


Figure 4. Overview map using Santiago data: (a) Time range 5:00–9:00 hrs. (b) Time range 5:00–9:00 hrs, after selecting hexagons in the communes of Santiago, Providencia, Las Condes, and Vitacura. (c) Time range 17:00–21:00 hrs, with the same hexagon selection.

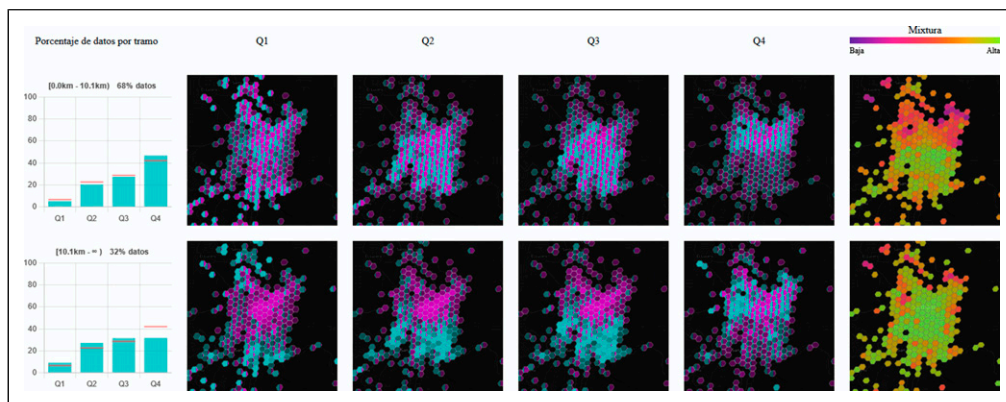


Figure 5. Visualization matrix panel with time range from 5:00 to 9:00 hours and a distance threshold of 10 km.

socioeconomic groups (Q1–Q3), which maintained more consistent participation across different distance ranges. This finding highlights that individuals of high socioeconomic status tend to make predominantly shorter trips, while other groups distribute their mobility more evenly between short and long distances. This difference likely reflects the spatial organization of Santiago, where high-income residents tend to live closer to employment centers and key urban amenities, allowing them to satisfy their mobility needs within shorter distances.

Observing the opacity of the hexagonal glyphs in both rows, we noticed that in group Q4, the glyphs located in the northern part of the city were more opaque compared to those in the southern part, while this marked difference was not observed in the other groups. This suggests that trips made by Q4 individuals are concentrated in the northern part of the city, consistent with previous observations indicating that high-class individuals tend to reside in the northeastern zone of the city (Toro and Orozco, 2018).

In the row corresponding to trips greater than 10 km, we observed that for groups Q2 and Q3, the hexagonal glyphs in the southern part of the city were predominantly cyan, while those in the northern part were predominantly magenta. This indicates that trips between 5:00 and 9:00 hours by individuals belonging to these groups typically go from south to north. This trend was also visible in the Q1 map, although less markedly, possibly due to the smaller amount of data available for this group. For group Q4, this south-to-north travel trend was not observed. However, we noted that origins tended to be located at the eastern and western extremes of the city, with destinations in the central area, consistent with our earlier discussion about employment activity centers.

Regarding the mixture maps, we observed that for trips greater than 10 km, the degree of mixture was relatively uniform, with lime color (high mixture) predominating in most of the city. In contrast, for trips less than 10 km, the northern zone of the city showed a lower degree of mixture. This may indicate that the travel experience for long-distance movements involves people from different socioeconomic levels, while short trips in the northern zone are more homogeneous in terms of the socioeconomic level of travelers. These patterns have direct implications for urban planning and equity. The concentration of high-income groups in short-distance mobility suggests unequal access to urban amenities, with lower-income residents facing longer commutes. The south-to-north morning flow pattern indicates a spatial mismatch between affordable housing and employment opportunities. Such insights can inform policies for transit investment prioritization, affordable housing location, and the decentralization of employment centers to create more equitable cities.

These findings demonstrate how MobilityHexVis can be used to uncover patterns of urban mobility across different socioeconomic groups and distance ranges. The ability to visualize and compare these patterns simultaneously is particularly valuable for understanding the socio-spatial dynamics of the city and identifying areas where interventions might be needed to promote greater social integration.

Pilot study

To evaluate the usability and effectiveness of MobilityHexVis, we conducted a preliminary pilot study with six participants who had experience with geospatial tools. The participants, all with backgrounds in computer engineering and experience in geographic information systems or data analytics software, were asked to perform a series of tasks designed to test the platform's functionality and gather qualitative feedback on the user experience.

Study design

The study was structured around four key user tasks that correspond to different levels of prior knowledge and interactive search approaches:

1. Open exploration: Participants were asked to adjust time ranges (5:00-9:00 and 17:00-21:00) to identify high-activity zones and changes in origin-destination dynamics without specific guidance.
2. Locating a known objective: Users needed to find areas with interesting mobility flows without knowing their exact location, using filtering capabilities.
3. Exploring a known area: Participants were directed to analyze specific geographic areas to understand how mobility flows vary across socioeconomic groups or time periods.
4. Direct querying: Users selected specific hexagons to obtain detailed information about origin and destination flows, including proportions and temporal dynamics.

Data was collected through an online survey implemented in Google Forms, which included an introduction explaining the platform's functionality, three task blocks with open-ended questions, and final sections about comprehension of hexagonal glyphs, design perception, improvement suggestions, and difficulties encountered during interaction.

Findings

Given the small sample size, we adopted a qualitative and exploratory approach, identifying initial impressions and improvement suggestions rather than drawing generalizable conclusions. The responses were grouped thematically around key aspects of the user experience.

Regarding interface comprehension and learning curve, two participants mentioned initial confusion with the origin and destination glyphs but reported greater comfort after brief use. Most participants positively valued the filtering controls for adjusting time range and distance thresholds, though some suggested adding the ability to directly compare two time periods without changing the slider. For municipality selection and detailed query mode, participants generally found the municipality selection useful, but some requested a way to highlight contours rather than filling the hexagons to avoid obscuring origin/destination information.

All participants agreed that the small multiples arrangement facilitates comparison across socioeconomic levels, though some expressed a desire for greater control to view multiple time periods simultaneously. Several participants specifically noted that the ability to compare

socioeconomic patterns side-by-side revealed insights that would be difficult to identify through sequential viewing or aggregated visualizations. Suggestions for improvement included adding the ability to download data, incorporating additional information layers, and providing a more guided tutorial.

In general, participants highlighted the clarity with which the platform illustrates mobility patterns across different times and socioeconomic groups. They appreciated the ability to filter by distance and municipality to focus on particular zones. Several agreed that the presentation of results through hexagonal glyphs and differentiated colors contributes to a relatively quick understanding of origin and destination distribution, provided clear explanations are offered about the meaning of the color splits.

One participant noted: ‘The hexagonal visualization makes it easier to identify patterns compared to traditional flow maps, especially when comparing across socioeconomic groups’. Another commented: ‘Being able to see all four socioeconomic groups simultaneously provides insights that would be difficult to notice otherwise’.

The pilot study confirmed the system’s usability and provided direction for future refinements, particularly regarding interface learning curve and temporal comparison features.

Conclusion

This paper presented MobilityHexVis, an interactive visualization platform for analyzing urban mobility flows across socioeconomic groups using mobile phone data. Our approach combines small multiples visualization with hexagonal glyphs to enable simultaneous comparison of mobility patterns among different socioeconomic groups and distance ranges.

Our case study in Santiago demonstrated the platform’s analytical capabilities by revealing that high socioeconomic status individuals predominantly make shorter trips compared to other groups, and by identifying distinct south-to-north travel patterns among middle socioeconomic groups during morning hours. The pilot study with domain experts confirmed the effectiveness of our visualization approach while identifying opportunities to improve the interface’s learning curve.

MobilityHexVis addresses a specific gap in existing tools by enabling simultaneous comparison of mobility patterns across socioeconomic groups. While individual components build on established techniques, their combination and application to socioeconomic mobility analysis represents a contribution that enables analyses previously impractical with existing tools.

We acknowledge limitations including the uneven distribution of base transceiver stations affecting spatial resolution, the simplified four-category socioeconomic classification, and the lack of information about transportation modes and trip purposes. Future work could address these by integrating complementary data sources and expanding the platform to include different temporal contexts and additional cities. Next iterations could implement pivot-table inspired functionality, allowing users to dynamically swap analysis dimensions. The modular architecture and Protocol Buffer data format support such extensions without requiring fundamental system redesign.

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Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This project was funded by the Chilean Agency of Research and Development, through a

FONDECYT Regular Grant # 1211490. This work was supported by National Center for Artificial Intelligence CENIA FB210017, Basal ANID and Fondo de Instalación FCFM Universidad de Chile.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

The Telefónica Movistar mobile phone records have been obtained directly from the mobile phone operator through an agreement between Universidad Austral de Chile and Telefónica R&D. This mobile phone operator retains ownership of these data and imposes standard provisions to their sharing and access which guarantee privacy. Anonymized datasets are available from Telefónica R&D Chile for researchers who meet the criteria for access to confidential data. The other datasets used in this study are publicly available.

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